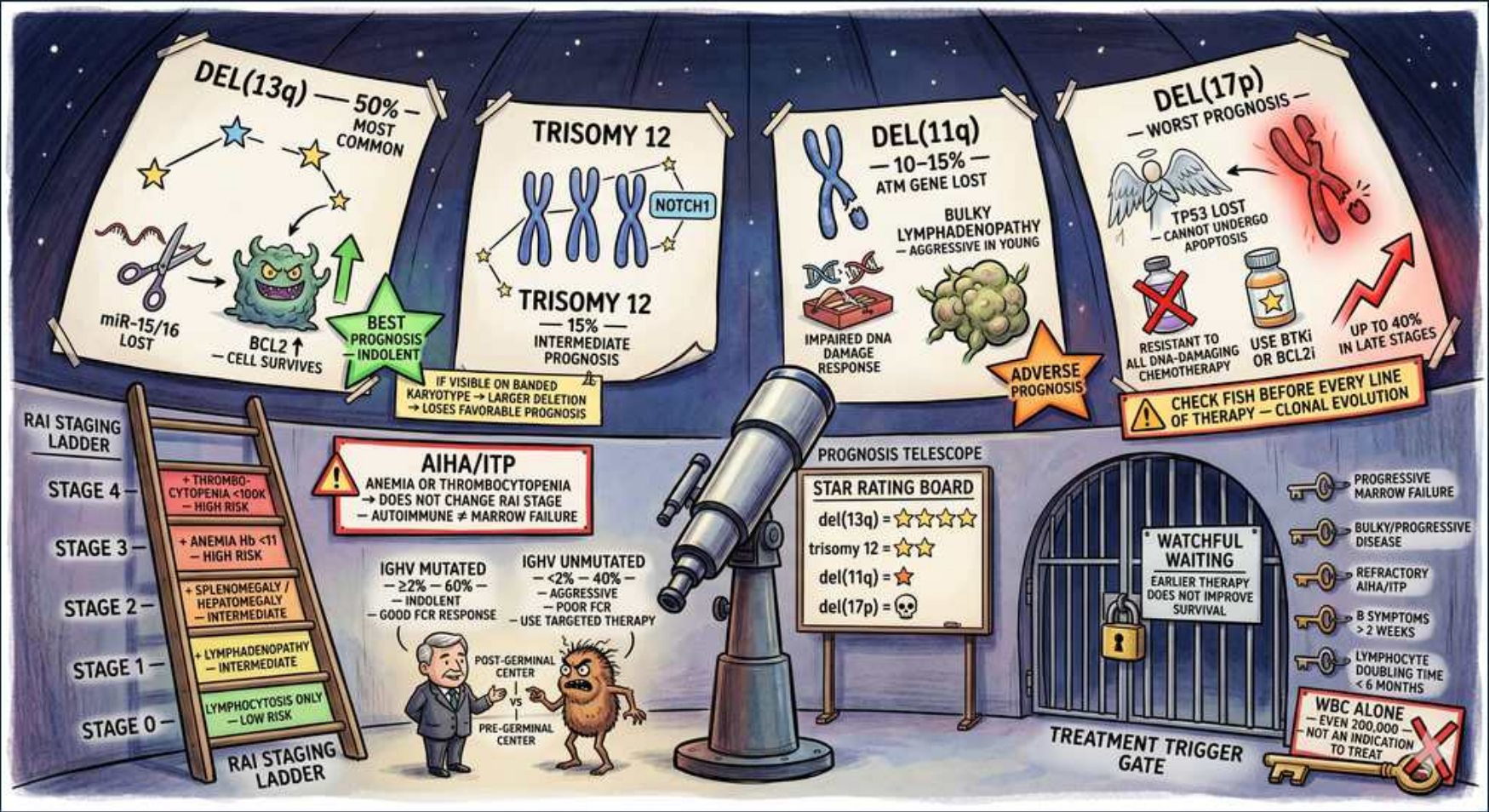


CLL Prognosis: FISH, IGHV, Rai Staging & Treatment Triggers



1. del(13q) — best prognosis

WHAT YOU SEE

Scissors panel: del(13q), '50% most common', best prognosis, indolent

WHAT IT ENCODES

Isolated del(13q) is the most common (~50%) cytogenetic abnormality and carries the BEST prognosis. miR-15/16 loss removes BCL2 suppression -> BCL2 up -> cell survives but disease is indolent.

BOARD TRAP

del(13q) is favorable; do NOT treat asymptomatic del(13q) CLL — it's the lowest-risk lesion, opposite of del(17p).

2. Trisomy 12 — intermediate

WHAT YOU SEE

Trisomy 12 chromosome panel with NOTCH1, ~15%, intermediate prognosis

WHAT IT ENCODES

Trisomy 12 (~15%) confers intermediate prognosis and associates with NOTCH1 mutations and atypical morphology/immunophenotype.

BOARD TRAP

Trisomy 12 is intermediate, not favorable — and NOTCH1 mutation adds adverse risk and predicts poor benefit from anti-CD20 antibodies.

3. del(11q) — ATM loss, bulky nodes

WHAT YOU SEE

del(11q) panel, 10-15%, ATM gene loss, bulky lymphadenopathy, aggressive in young

WHAT IT ENCODES

del(11q) (10-15%) deletes the ATM gene -> impaired DNA damage response -> bulky lymphadenopathy and aggressive disease, especially in younger patients. Historically resistant to chemo.

BOARD TRAP

del(11q) means impaired DNA repair (ATM), so it predicts poor chemoimmunotherapy response — but responds well to BTK inhibitors (the modern fix).

4. del(17p) — worst prognosis

WHAT YOU SEE

del(17p) panel, TP53 lost, 'cannot undergo apoptosis', up to 40% in late stages

WHAT IT ENCODES

del(17p)/TP53 loss is the WORST prognostic marker — cells cannot undergo p53-mediated apoptosis, so chemoimmunotherapy fails. Up to 30-40% in late/relapsed disease.

BOARD TRAP

del(17p)/TP53-mutated CLL must NOT receive chemoimmunotherapy (FCR) — it is refractory; use BTKi or venetoclax (targeted agents) instead.

5. Check FISH before therapy

WHAT YOU SEE

Sign: 'check FISH before every line of therapy — clonal evolution'

WHAT IT ENCODES

FISH (and TP53 sequencing) must be repeated before EACH line of therapy because clonal evolution can acquire del(17p)/TP53 over time, changing treatment choice.

BOARD TRAP

A prior normal FISH does not stay valid — clonal evolution means you re-test for del(17p)/TP53 before every treatment line.

6. IGHV mutated vs unmutated

WHAT YOU SEE

AIHA/ITP box adjacent to IGHV mutated (good) vs unmutated (poor) panel

WHAT IT ENCODES

IGHV MUTATED (>2% from germline) = good prognosis, indolent. IGHV UNMUTATED (<2%) = poor prognosis, more aggressive. Counterintuitive: 'mutated' is the GOOD one.

BOARD TRAP

Counterintuitive marker — IGHV MUTATED = favorable, UNMUTATED = unfavorable. Students often reverse this. Unmutated also predicts shorter FCR durability.

7. AIHA / autoimmune cytopenia

WHAT YOU SEE

Box: AIHA — anemia/thrombocytopenia, autoimmune marrow failure

WHAT IT ENCODES

Autoimmune hemolytic anemia (and immune-mediated low platelets) in CLL is an autoimmune phenomenon, NOT marrow infiltration — it does NOT upstage Rai/Binet and is treated with steroids.

BOARD TRAP

Immune cytopenias do NOT count as advanced-stage (Rai III/IV) disease — autoimmune anemia is treated with steroids, not as an indication to start CLL-directed chemo.

8. Rai staging ladder (0-4)

WHAT YOU SEE

Vertical ladder, stages 0-4, low/intermediate/high risk

WHAT IT ENCODES

Rai: Stage 0 = lymphocytosis only (low); I-II = nodes/spleen/liver (intermediate); III = anemia (Hgb<11); IV = thrombocytopenia (<100k) (high risk). Cytopenia must be from marrow infiltration.

BOARD TRAP

Rai III/IV requires INFILTRATIVE cytopenia (marrow replacement); autoimmune cytopenia does not qualify as high-stage.

9. Prognosis telescope (star ratings)

WHAT YOU SEE

Telescope and star-rating board: del(13q) high stars down to del(17p) low stars

WHAT IT ENCODES

Integrated prognosis: del(13q) best, trisomy 12 intermediate, del(11q) adverse, del(17p) worst — visualized as descending star ratings predicting survival.

BOARD TRAP

Stacking of adverse markers (del17p + unmutated IGHV + complex karyotype) compounds risk — single favorable FISH can be overridden by TP53 status.

10. Watchful waiting (early stage)

WHAT YOU SEE

'Watchful waiting' gate; earlier treatment does not improve survival

WHAT IT ENCODES

Asymptomatic early-stage CLL is observed — treating early does NOT improve overall survival. Treatment begins only when triggers are met.

BOARD TRAP

Do NOT treat asymptomatic Rai 0-II CLL just because counts are abnormal; early treatment offers no survival benefit (classic exam trap).

11. Treatment trigger gate

WHAT YOU SEE

Gate with triggers: progressive marrow failure, bulky/progressive adenopathy, refractory autoimmune cytopenia, B symptoms, doubling time <6 months

WHAT IT ENCODES

iwCLL treatment indications: progressive marrow failure (worsening anemia/thrombocytopenia), massive/progressive/symptomatic splenomegaly or nodes, lymphocyte doubling time <6 months, steroid-refractory autoimmune cytopenia, or B symptoms.

BOARD TRAP

A high WBC alone is NOT a treatment indication — absolute lymphocyte count, even very high, does not by itself trigger therapy; you need a functional/symptomatic trigger.

12. WBC alone is not an indication

WHAT YOU SEE

Banner: 'WBC alone — even 200,000 — not an indication to treat'

WHAT IT ENCODES

Unlike acute leukemia, very high lymphocyte counts in CLL rarely cause leukostasis, so an isolated elevated WBC does not mandate treatment.

BOARD TRAP

CLL does NOT cause leukostasis the way AML/CML blast crisis does — don't treat purely for a high count; treat for the iwCLL triggers.
